IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for High Purity Germanium Radiation Detectors and Associated Electronics

PROCEDURE Nº MRNI-507
REV. D0

Instituto Nacional de Investigaciones Nucleares
MÉXICO

DECEMBER 2008

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CONTENT

PAGE

1.- OBJECTIVE AND SCOPE. 4

1.1.- Objective. 4
1.2.- Scope. 4

2.- NOTATION AND DEFINITIONS 4

2.1.- Notation. 4
2.2.- Definitions. 4

3.- DEVELOPMENT 5

3.1.- Introduction 5
3.2.- Test Instruments 6
3.3.- Radioactive source 7
3.4.- Test of the detector. 7
3.5.- Test of the preamplifier. 9
3.6.- Test with an amplifier 12
3.7.- Test with a multichannel analyzer system. 14
3.8.- Measurement of the relative efficiency of the detector. 16

4.- ADMINISTRATION OF THE REPORTS 17

4.1.- Numbering of reports 17
4.2.- Personnel 17
4.3.- Technical Report 17

5.- ACTION IN CASE OF NON CONFORMITIES 17

5.1.- Technical Report 17
5.2.- Labelling 17

6.- RESPONSIBILITIES 17

6.1.- Manager 17
6.2.- Personnel 18

7.- BIBLIOGRAPHY. 18

8.- ANNEXES 18

Annex I Flow chart 20
Annex II Test report 24

1.- OBJETIVE AND SCOPE

1.1.- Objective.

This test procedure establishes the principal activities to verify the performance of gamma spectrometers based on High Purity Germanium radiation detectors (HpGe).

1.2.- Scope.

This procedure can be applied in different electronic laboratories to verify gamma spectrometers based on cooled Hp-Ge detectors.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

FWHM	Full Width at Half Maximum of a peak in a spectrum
FWTM	Full Width at one Tenth Maximum of a peak in a spectrum
ININ	National Institute of Nuclear Research
MCA	Multichannel Analyzer
NIM	Nuclear Instruments Modules

2.2.- Definitions

2.2.1 Bias voltage for semiconductor radiation detector
The reverse voltage applied to a detector to produce the electric field and sweep out the signal charge generated inside the detector.

2.2.2 Cryostat
Assembly that contains the crystal detector, keeps it in vacuum and provides a thermal contact between the crystal and the liquid nitrogen.

2.2.3 Detector
A Device that converts the energy of a photon or an incident particle in an electric pulse, typically a current or charge pulse.

2.2.4 Leakage currents in a semiconductor detector
Low current that is present when the detector is reverse biased and there is not a radiation field.

2.2.5 Charge sensitive preamplifier
A device for coupling impedances, it amplifies and integrates the charge that is produced in the detector.

2.2.6 Spectrometry
Measurement of the energy distribution of the incident radiation.

3.- DEVELOPMENT.

3.1 Introduction

Semiconductor gamma spectrometers consist of a semiconductor radiation detector assembly and signal processing electronics interfaced to a pulse height analyzer and a computer. The detector is a semiconductor crystal between two conductor electrodes. A potential difference is established between the electrodes thereby producing an electric field in the semiconductor. When a γ-ray photon enters the semiconductor it produces free charge carriers in the crystal, the number of which is proportional to the energy lost by the γ-ray photon. The charge motion resulting from the influence of the electric field produces an induced current pulse in the external circuit. The integrated current pulse is proportional to the energy lost by the γ-ray photon.

The pulses are routed to a multichannel pulse-height analyzer (MCA) where they are sorted and stored according to the amplitude distribution to produce a pulse-height graph that corresponds to the gamma energy spectrum. The MCA may be a dedicated instrument or an analog-to-digital converter (ADC) interfaced to a computer (see Fig. 1).

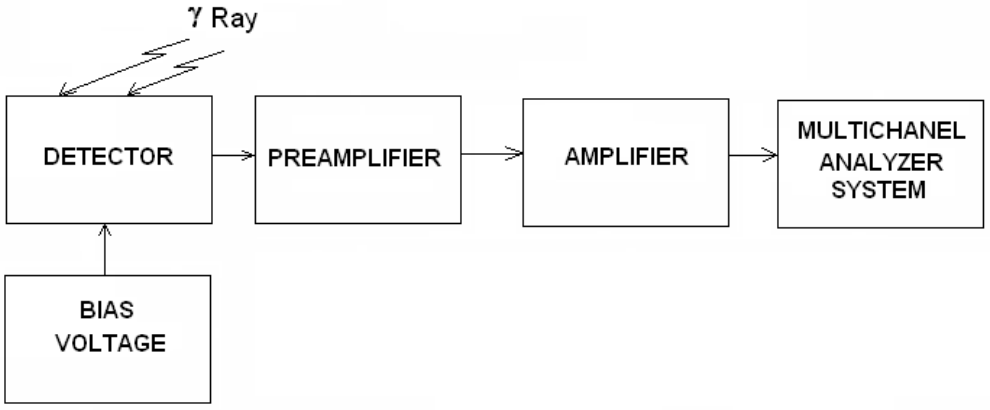


Fig. 1.- Blocks diagram of gamma spectroscopy system

The semiconductor material most frequently used for gamma spectrometers is germanium; it does not have sufficient high resistivity to withstand large electric fields without excessive leakage currents. It is therefore necessary to use special techniques to limit the current flow through the semiconductor device. This is usually accomplished by utilizing the space charge region of a reverse biased diode junction as the detector-sensitive volume.

The problem of noise in the detector and in the preamplifier circuit is significant, consequently, both the semiconductor detector and the first stage of the preamplifier are usually operated at very low temperature to reduce the thermal noise.

Energy resolution is one of the most important characteristics of spectrometers since it sets the limit on the ability to resolve closely spaced lines in the spectrum. Other important parameters are count-rate capability, gain stability, and detector size (efficiency).

Electronic stability is an important variable that includes not only the stability of the preamplifier, but also that of other parts of the system, such as the main amplifier and MCA. Effects of high count rate such as pulse pile-up and dc level shifts are also important. Every time an event produces a pulse in the amplifying equipment, the dc levels throughout the system are perturbed and take some time to return to their original values. If another event occurs within this time interval, its effective output pulse height may be altered, thereby contributing to spectral distortion.

The sequence of activities to verify the characteristics of gamma spectrometers based on an HpGe detector are described in the next paragraphs, also a flux graph is included in Annex I.

3.2 Test Instruments.

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

3.2.1 NIM Beam.

A NIM beam holds all the NIM modules and provides the necessary voltages for its operation.

3.2.2 Detector Bias Power Supply.

Generally it is a high voltage power supply with enough current capability to feed the detector without any loose in regulation, regulation less than 0.01 %, the power supply must have a good stability. The ripple should be less than 100 mV.

3.2.3 Pulse Amplifier.

A spectroscopy amplifier with shaping times from 3μs to 12 μs is required.

3.2.4 Oscilloscope

Use an analog or digital oscilloscope .

3.2.5 Counter or Scaler

The number of counts per time unit is measured with a counter or scaler for nuclear pulses it generally includes a voltage discriminator to block low amplitude noise pulses. The counter measures positive pulses.

3.2.6 Rate Meter

The average number of counts per unit of time is measured with a rate meter for nuclear pulses. The rate meter measures positive pulses.

3.2.7. Multichannel Analyzer, MCA

A MCA with at least 1024 channels is needed to get the energy spectrum. Consider that the analog input pulse must be positive.

3.2.8. Scintillation detector

A 3" × 3" NaI(Tl) detector with its high voltage power supply and preamplifier is required for the measurement of the relative efficiency.

3.3 Radioactive source

It is necessary to have radioactive sources of ^{137}Cs and ^{60}Co, with an activity of approximately 37000 Bq (1 μCi).

3.4 Test of the detector.

This test determines the detector condition.

3.4.1 Review the detector manual and determine the detector parameters and configuration (operating voltage and polarity, type of preamplifier, type of semiconductor, etc.).

3.4.2 Make a visual inspection of the cryostat to verify the integrity of the assembly, check that the cryostat is not struck, if the detector has a beryllium window, it has to be in good conditions; it must not be broken (see Fig. 2).

3.4.3 If the cryostat is in good conditions, put it into liquid nitrogen and wait for at least 6 hours (see Fig. 3). Observe the detector carefully, in case that it gets wet or frozen (see Fig. 4), probably there is a vacuum problem and it has to be repaired, If there is not sign of failure continue with the test.



Fig 2.- Semiconductor detector with the beryllium window damaged



Fig 3.- Semiconductor detector in liquid nitrogen



Fig 4.- Semiconductor detector with a vacuum problem

3.5 Test of the preamplifier

3.5.1 Check the DC voltage values in the NIM bin, you must measure +24 V, -24 V, +12 V, -12 V and in some cases also +6 V and -6 V.

Register the measured values in the test report, Annex II.

3.5.2 Generally the detector has included a preamplifier very close to the output port of the detector. Determine the type of preamplifier that the detector has, there are only two possibilities: a) Resistive feedback b) Transistor Reset Preamplifier.

A Transistor Reset preamplifier also has an inhibit output; this could help to identify the type of preamplifier.

Register this information in the test report, Annex II

3.5.3 Connect the detector to the electronic modules as described in the next steeps (see Fig. 5).

3.5.4 Connect the preamplifier power cable to the power socket, DB9 connector, on the rear panel of the main amplifier

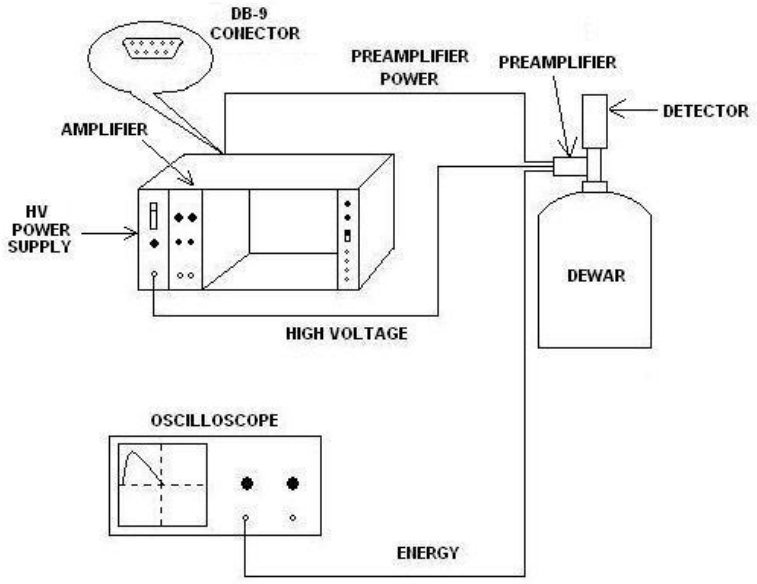


Fig. 5.- Connection of a HpGe detector to see the output signal from the preamplifier

3.5.5 Check the polarity of the high voltage or bias supply (HV) which is indicated on both the detector label and in the data sheet supplied with the detector; register it in the test report Annex II and verify that the high voltage power supply is set to the correct polarity.

3.5.6 Check that the HV power supply is turned down completely to zero and the switch which is in the front panel is set to off.

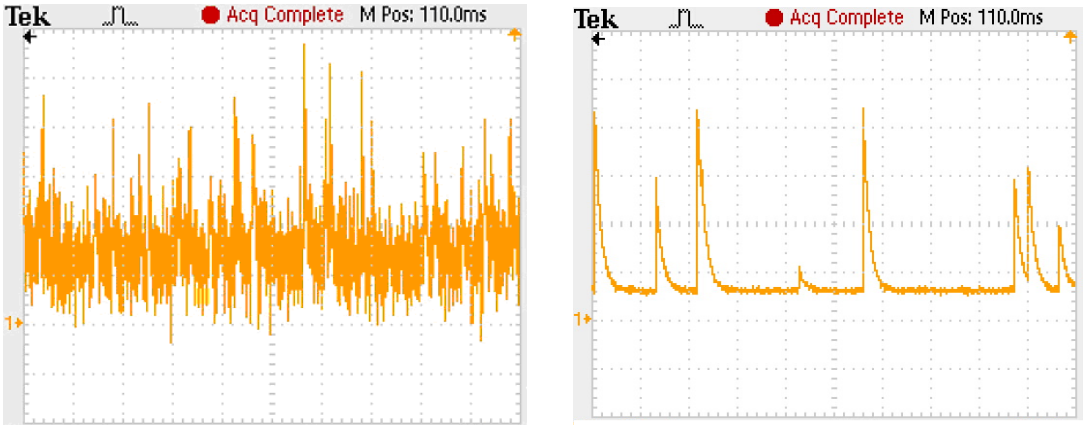
CAUTION: Never connect or disconnect the detector when the high voltage power supply is on.

3.5.7 Using a HV coaxial cable, connect the output of the HV power supply to the bias input of the preamplifier.

3.5.8 Place a radioactive source (^{137}Cs or ^{60}Co) near the detector

3.5.9 Connect the output of the preamplifier to the (DC) input of the oscilloscope, and adjust the oscilloscope to auto trigger (DC) with a sweep of 100 μms/division and 50 mV/division.

Some pulses will appear in the oscilloscope screen at the preamplifier output, even without the bias voltage. See Fig. 6.a. The pulses could be positive or negative depending of the design of the preamplifier.



a) b)

Fig. 6.- Positive signals at the output of a resistive feedback preamplifier: a) without the bias voltage applied to the detector; b) with the bias voltage. The settings of the oscilloscope are: 50 mV/div. and 100 μs/div.

3.5.10 Find the value of the appropriate high voltage or bias supply (HV) for the detector, which is indicated on both the detector label and in the data sheet supplied with the detector.

3.5.11 Turn on the HV power supply and increase the voltage at 100 V.

3.5.12 Look at the waveform in the oscilloscope, if you have a preamplifier with resistive feedback you will observe that the trace will go up or down every time the high voltage changes, wait until the trace reaches the base line, also some random pulses will appear, the shape of that pulses is similar to the ones shown in Fig. 6.b, notice that the noise reduces every time the applied voltage is increased at the begining. If there is not a signal, the preamplifier is not working properly and it has to be reviewed and repaired.

3.5.13 On the other hand, if the preamplifier has an transistor reset preamplifier, you will see a signal similar to the one show in Fig. 7, with a bias voltage of around 100 V.
If there is not a signal, the preamplifier is not working properly and it has to be reviewed and repaired.

3.5.14 Increase the high voltage in steps of 100 V every 10 seconds until the detector reaches the operating voltage.

3.5.15 Draw the obtained signal in the test report, Annex II.

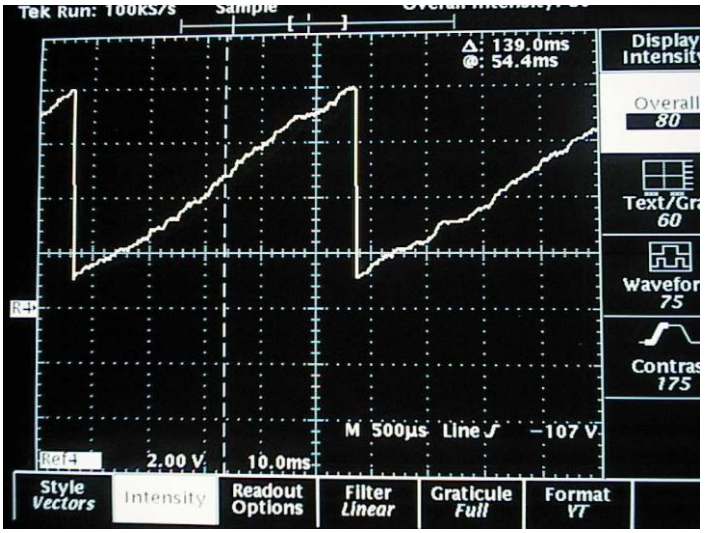


Fig. 7.- Signal from a transistor reset preamplifier. The settings of the oscilloscope are: 2 V/div. and 10 ms/div.

3.6 Test with an amplifier.

3.6.1 Connect the preamplifier output to the main amplifier input with a BNC coaxial cable (see Fig. 1); adjust the appropriate input polarity on the amplifier according with the pulse observed in paragraph 3.5.12 or 3.5.13.

3.6.2 Adjust the main amplifier as follows:

Coarse gain:	20
Fine gain:	full
Shaping time:	4 μS or larger
Baseline Restorer:	Auto or Low; ASYM
Threshold:	Auto
Pole Zero (P/Z):	Mid range if the preamplifier uses resistive feedback, Completely anticlockwise for a transistor reset preamplifier.

3.6.3 Connect the main amplifier output to an oscilloscope input as shown in Fig. 8 and place a radioactive source (^{137}Cs, or ^{60}Co).

3.6.4 Adjust the amplifier coarse gain and fine gain to get a signal of about 5 Volts in amplitude.

3.6.5 You must observe a signal as the one shown in Fig. 9.

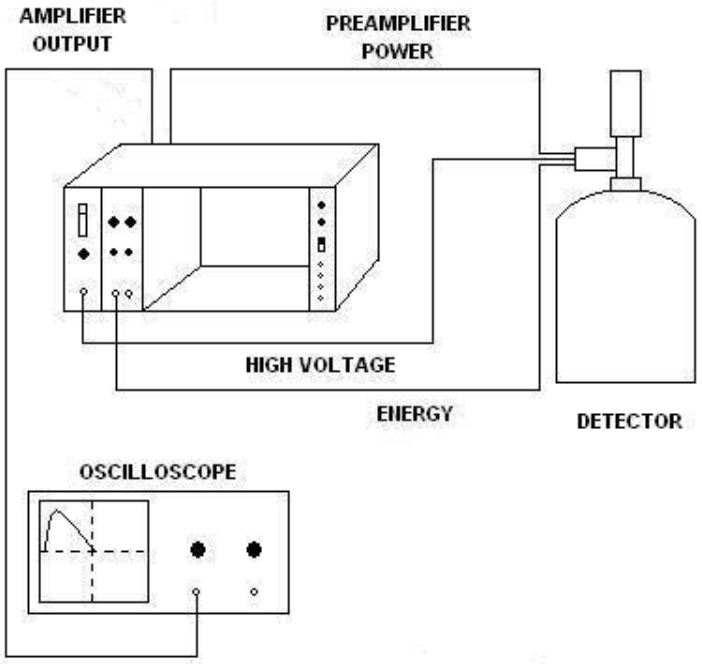


Fig. 8.- Connections to measure the output signal of the main amplifier

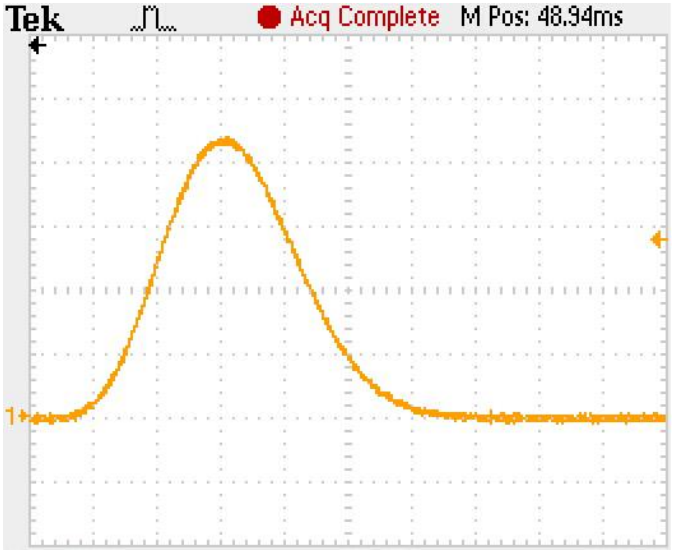


Fig 9.- Amplifier output signal. The settings of the oscilloscope are: 1 V/div. and 5 μs/div.

3.6.6 Register the value of the coarse gain and fine gain and draw the output signal in the test report, Annex II

3.6.7 If you have a resistive preamplifier and the output signal has an overshoot or undershoot as shows in Fig. 10, you must adjust the pole-zero in the main amplifier to correct the pulse.

3.6.8 If you do not observe any signal, probably your amplifier is not working properly and it has to be reviewed and repaired.

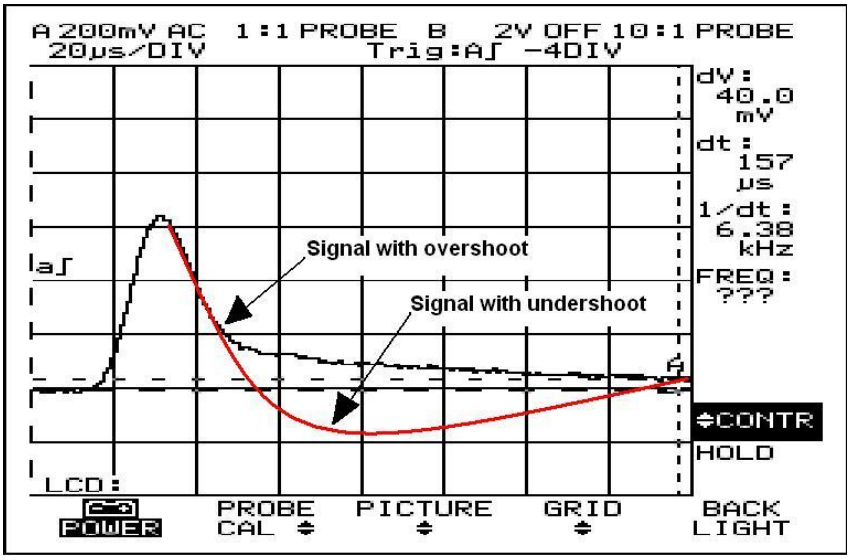


Fig. 10.- Amplifier output signal without proper compensation. The settings of the oscilloscope are: 200 mV/div. and 20 µs/div.

3.7 Test with a MCA.

3.7.1 If the signal at the amplifier output is right, connect it to the MCA input.

3.7.2 Place a ^{60}Co radioactive source which emits gamma rays of 1173.2 keV and 1332.5 keV. Place the source at an adequate distance to obtain a pulse repetition rate not greater than 2 000 cps to minimize dead time effects.

3.7.3 Acquire an energy spectrum for 10 minutes or the enough time to get a net count in the peaks of around 10 000 counts to minimize the relative uncertainty of the results (< 1%). Use the peak energies to calibrate the MCA as shows in Fig. 11.

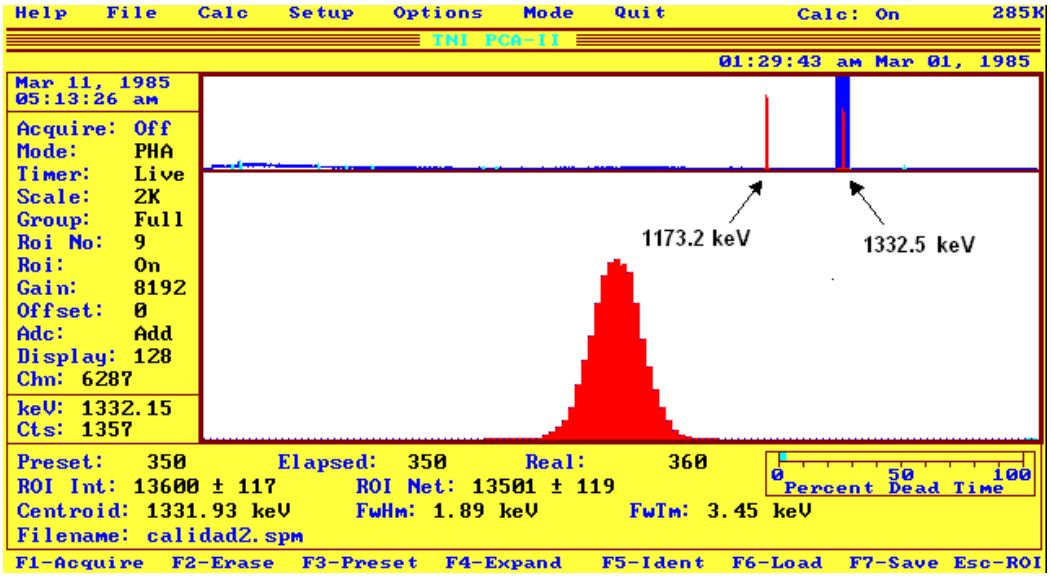


Fig. 11.- Energy spectrum of ^{60}Co

3.7.4 Make a calibration of the MCA using the routine for that purpose or obtain the Energy calibration equation for the Spectrometry System:

$$y = m\,x + b$$

where: y = energy value m = energy per channel
x = channel value b = energy at channel 0

3.7.5 If you made the calibration process using the software routine, then measure the FWHM in the peak of 1332.5 keV for ^{60}Co, using the MCA (see Fig. 11).

3.7.6 On the other hand, if you made the calibration manually and you know the energy per channel, *m*, get the number of channels at the FWHM and then, convert them to an energy value.

3.7.7 In both cases the result must be around 1.9 keV, if you get a higher value, reduce very carefully the high voltage (steps of 100 V every 10 seconds) until you have zero volts, turn off the low voltage power supply, dry carefully the preamplifier with warm air and check that all the BNC coaxial cables are in good conditions.

3.7.8 Calculate the form factor (FWTM/FWHM) and register it in the test report, Annex II, the ideal value should be less than 1.9.

3.8 Measurement of the relative efficiency of the detector.

a) With any ^{60}Co radioactive source:

3.8.1 Set the spectroscopy system in the normal conditions of operation, including the operating voltage of the detector, place a ^{60}Co source at 25 cm from the end cap of the detector and just in its symmetry axe. The activity of the source must be such as that the counting rate be around 1000 cps to avoid pile up and dead time errors.

3.8.2 Make a region of interest around the 1332.5 keV peak of ^{60}Co and accumulate for a live time T1 enough to get a net area, C1, of more than 10 000 counts.

3.8.3 Repeat the former measurement with a calibrated spectroscopy system with a 3"x3" (NaI)Tl scintillation detector for the same time T1. The net area obtained around the 1332.5 keV peak will be C2.

3.8.4 The relative efficiency in per cent, E_{ff}, is calculate as:

$$E_{ff} = \frac{C1}{C2} \times 100$$

b) With a calibrated ^{60}Co radioactive source:

3.8.5 Set the spectroscopy system in the normal conditions of operation, including the operating voltage of the detector, place a ^{60}Co source with a certified activity A in disintegrations per second, at 25 cm from the end cap of the detector and just in its symmetry axe. The activity of the source must be such as that the counting rate be around 1000 cps to avoid pile up and dead time errors.

3.8.6 Make a region of interest around the 1332.5 keV peak of ^{60}Co and accumulate for a live time T1, in seconds, enough to get a net area, C1, of more than 10 000 counts.

3.8.3 The relative efficiency in per cent, E_{ff}, is calculate as:

$$E_{ff} = \frac{C1}{1.2 \times 10^{-3}\ A\ T1} \times 100$$

4.- ADMINISTRATION OF THE TECNICAL REPORTS

4.1 Numbering of the reports.

All the technical reports generated must have a unique and consecutive number.

4.2. Personnel.

All the operations in the detector and the associated instruments must be realized by qualified personnel.

4.3. Technical Report.

4.3.1 At the end of the diagnostic test of the spectrometry system, make a technical report, starting with the description of the detector, mark, model, serial number, it will indicate the operative conditions of the system and the results of the different tests, including the name of the person who made the tests.

4.3.2 All the technical reports must be classified and keep in a folder for future consult.

5.- ACTION IN CASE OF NON CONFORMITIES.

5.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

5.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

6.- RESPONSIBILITIES

6.1.- Manager

6.1.1 Verify that all the diagnostic operations are made according to this procedure.

6.1.2 Check the activities of the personnel.

6.2 Personnel

6.2.1 Be sure that all the test equipments are in operative conditions and calibrated.

6.2.2 Make the test of the detector, preamplifier, amplifier and MCA according to the procedures.

6.2.3 Write the reports of the entire test carried out.

6.2.4 Elaborate the technical report for the system.

6.2.5 Make a control register of the technical reports

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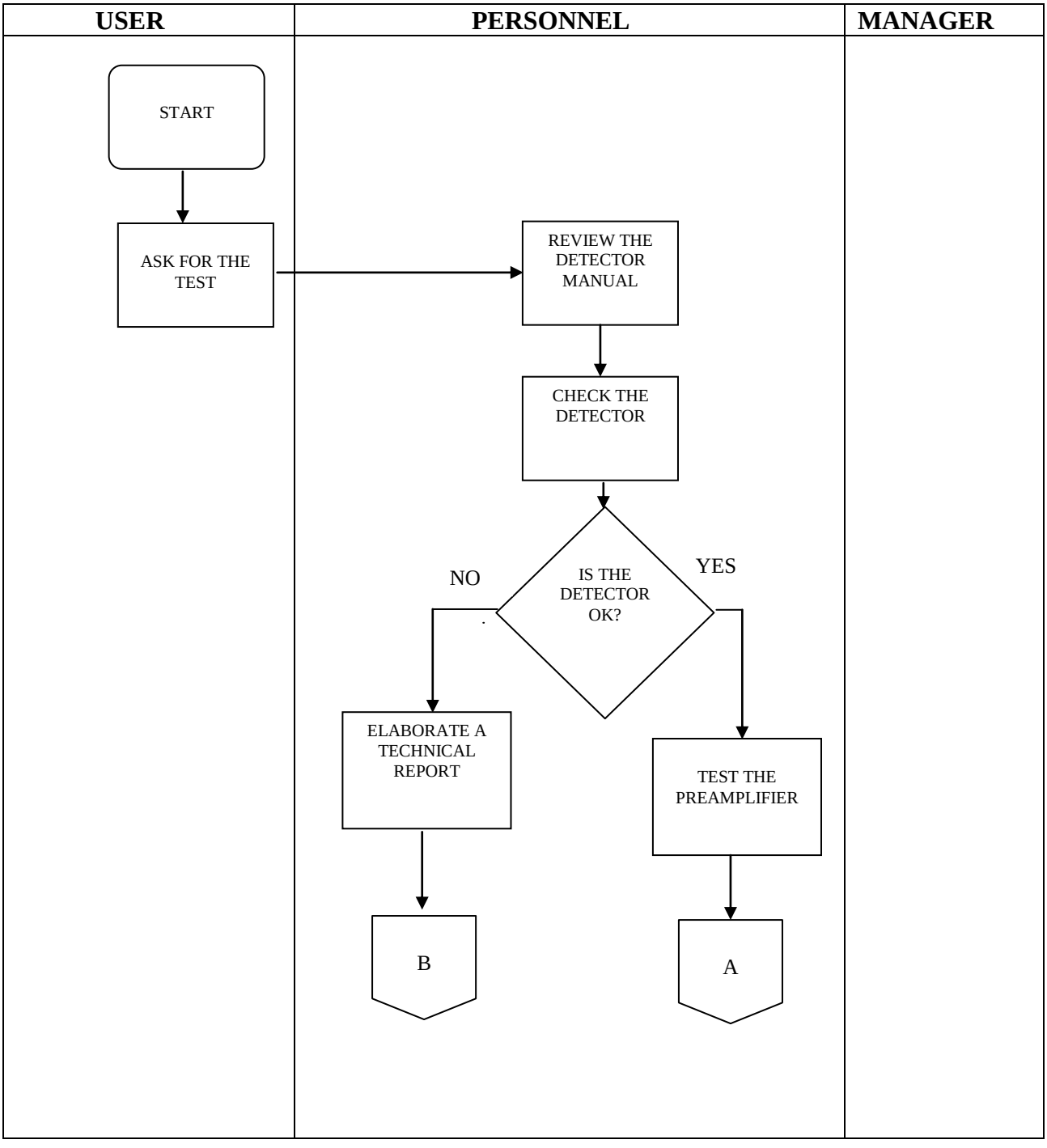
8.- ANNEXES

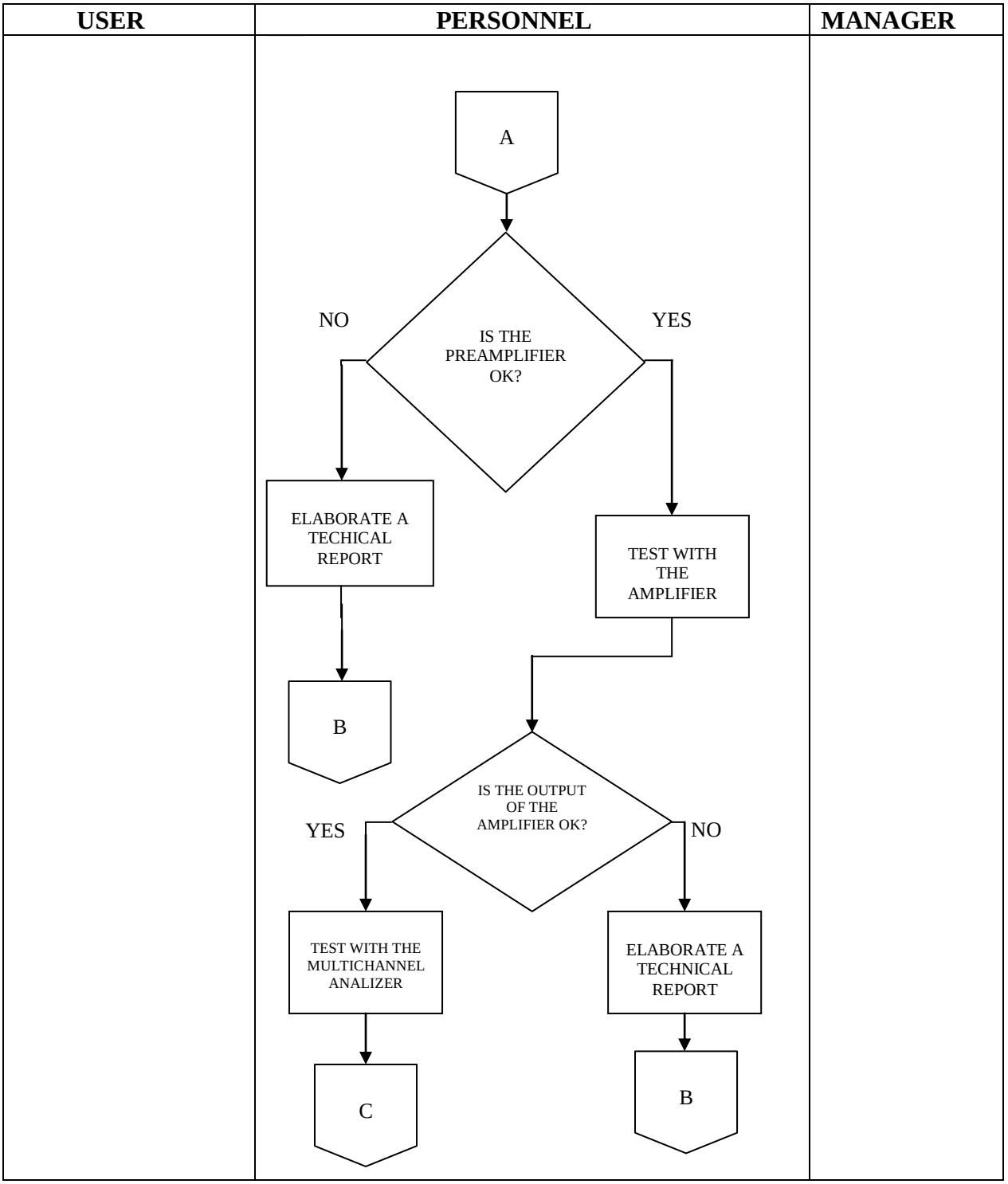
Annex I.- Flow chart

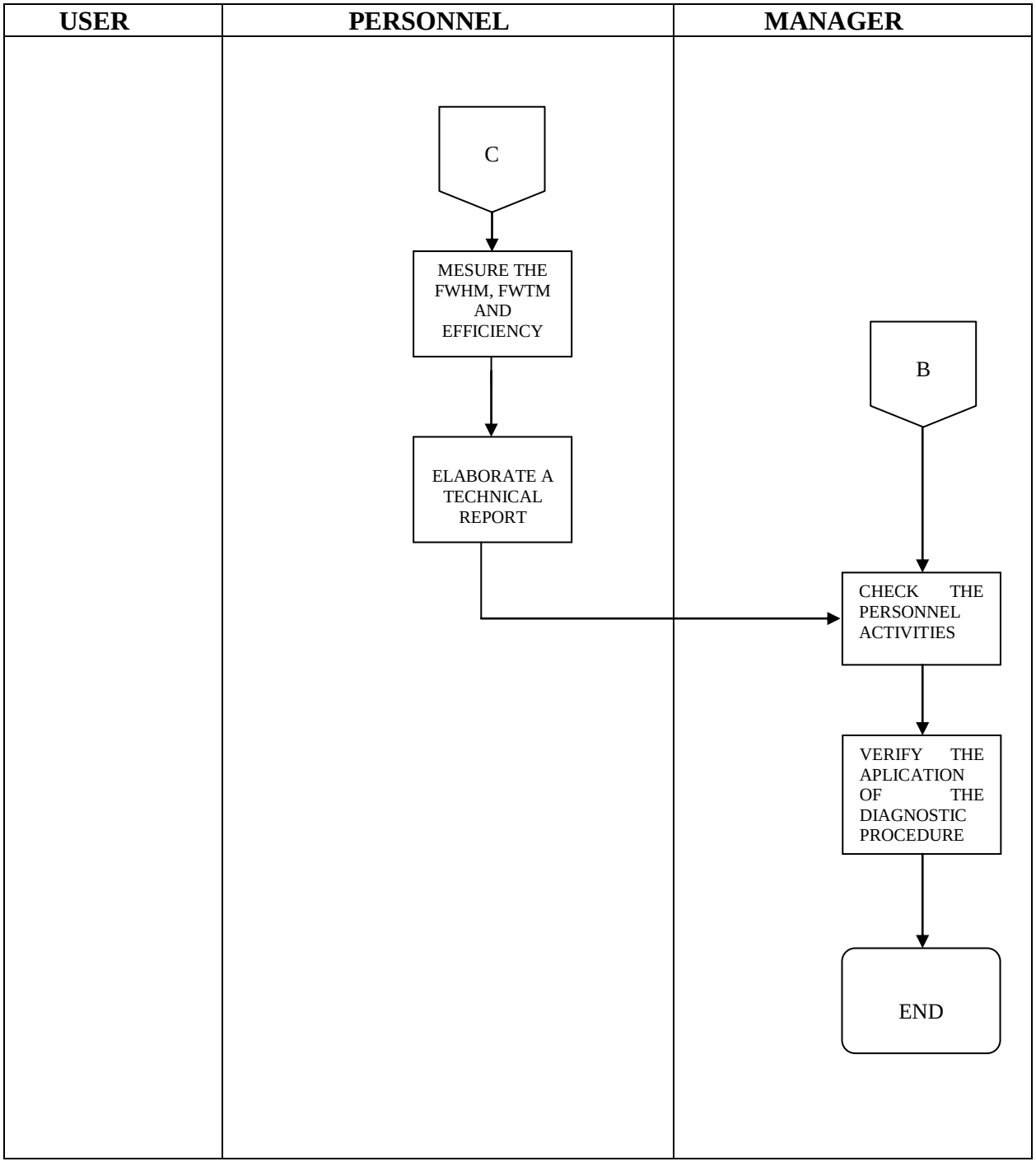
Annex II.- Test report

ANNEX I

Flow chart


USER
PERSONNEL
MANAGER
START
ASK FOR THE TEST
REVIEW THE DETECTOR MANUAL
CHECK THE DETECTOR
IS THE DETECTOR OK?
NO
YES
ELABORATE A TECHNICAL REPORT
TEST THE PREAMPLIFIER
B
A


USER
PERSONNEL
MANAGER
A
IS THE PREAMPLIFIER OK?
NO
YES
ELABORATE A TECHICAL REPORT
TEST WITH THE AMPLIFIER
B
IS THE OUTPUT OF THE AMPLIFIER OK?
YES
NO
TEST WITH THE MULTICHANNEL ANALIZER
ELABORATE A TECHNICAL REPORT
C
B



Annex I.- Flow graph

ANNEX II

Test report

TEST REPORT Nº HpGe-_____

HP-Ge DETECTOR

Mark: _____	**Model: _____**
Serial Nº: _____	**Window: _____**

Conditions

Cooling time: _____	***Bias voltage:* _____ *Polarity*: _____**
Type of preamplifier: Transistor reset_____ Resistive feedback_____	
Are the cryostat and the beryllium window Ok? Yes_____ No_____	***Is the cryostat wet or frozen after the cooling cycle?*** Yes_____ No_____
Voltages measured in the low voltage power supply. **+24 _____ -24 _____** **+12 _____ -12 _____** **+6 _____ -6 _____**	***Time constant of the amplifier: _____μS*** ***Base line restoration: _____*** ***Amplifier coarse gain: _____*** ***Amplifier fine gain: _____***
Draw the signal obtained in the preamplifier output.	***Draw the signal obtained in the amplifier output.***

Test equipment

Amplifier:

Mark: _____

Model: _____

Serial Nº: _____

High voltage power supply:

Mark: _____

Model: _____

Serial Nº: _____

Multichannel analyzer:

Mark: _____

Model: _____

Serial Nº: _____

Results

Resolution (FWHM): _____ measured in 1332.5 keV, peak of ^{60}Co radiation source.

Form factor of the pulse (FWTM/FWTM): _____ for 1332.5 keV.

Relative efficiency:_____% Radioactive source:_____

Diagnostic

Tested by: _____

Date: _____